



COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	9/01/2026	DATE SUBMITTED:	9/02/2026
NAME:	Aicon H. Keliste	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 6.1 Targeting Specific Nodes

1. PROCEDURE OUTPUT

Task 1:

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 site.yml
---
- hosts: all
  become: true
  tasks:

  - name: install apache and php for Ubuntu Servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
      when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    dnf:
      name:
        - httpd
        - php
      state: latest
      when: ansible_distribution == "CentOS"
```

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 hosts *
[web_servers]
192.168.56.103
192.168.56.104

[db_servers]
192.168.56.105

[file_servers]
192.168.56.106
```

GNU nano 6.2

site.yml *

```
---
- hosts: all
  become: true
  pre_tasks:

    - name: install updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  tasks:

    - name: install apache and php for Ubuntu Servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```



```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
Information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False
in ansible.cfg.
ok: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [install updates (CentOS)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.104]
ok: [192.168.56.103]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]
ok: [192.168.56.105]
ok: [192.168.56.103]
ok: [192.168.56.104]

TASK [install apache and php for Ubuntu Servers] *****
skipping: [192.168.56.105]
ok: [192.168.56.103]
ok: [192.168.56.104]
changed: [192.168.56.106]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY RECAP *****
192.168.56.103      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.104      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.105      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106      : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

keliste@Keliste-Workstation: ~/CPE212_KELISTE$
```



keliste@Keliste-Workstation: ~/CPE212_KELISTE

GNU nano 6.2

site.yml *

```
  state: latest
  update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
    when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    dnf:
      name: mariadb-server
      state: latest
      when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    apt:
      name: mariadb-server
      state: latest
      when: ansible_distribution == "Ubuntu"
```

```

keliste@Keliste-Workstation: ~/CPE212_KELISTE

ok: [192.168.56.103]
ok: [192.168.56.104]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]
ok: [192.168.56.105]
ok: [192.168.56.104]
ok: [192.168.56.103]
ok: [192.168.56.104]

TASK [install apache and php for Ubuntu Servers] *****
skipping: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.104]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
changed: [192.168.56.105]

TASK [Mariadb- Restarting/Enabling] *****
changed: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
skipping: [192.168.56.105]

PLAY RECAP *****
192.168.56.103      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.104      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.105      : ok=7    changed=2    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.106      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$

```

Centos - Systemctl status mariadb

```

[keliste-centos@KelisteMain CPE212_KELISTE]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-08-27 08:00:10 PST; 2min 18s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 100238 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 100262 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 100358 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 100345 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 8 (limit: 10502)
    Memory: 72.1M (peak: 107.0M)
       CPU: 476ms
    CGroup: /system.slice/mariadb.service
            └─100345 /usr/libexec/mariabdd --basedir=/usr

[keliste-centos@KelisteMain CPE212_KELISTE]$

```

Describe: The mariadb service is loaded and enabled from the package unit file, and shows active (running).

```

GNU nano 6.2                                                                    keliste@Keliste-Workstation: ~/CPE212_KELISTE
site.yml *
when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

- name: install mariadb package (CentOS)
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/Enabling"
  service:
    name: mariadb
    state: restarted
    enabled: true

- name: install mariadb package (Ubuntu)
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- hosts: file_servers
  become: true
  tasks:

- name: install samba package
  package:
    name: samba
    state: latest

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Localize
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To
  
```



COMPUTER
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```
keliste@Kelliste-Workstation: ~/CPE212_KELISTE
ok: [192.168.56.105]

TASK [Mariadb- Restarting/Enabling] *****
*****
changed: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
*****
skipping: [192.168.56.105]

PLAY [file_servers] *****
*****

TASK [Gathering Facts] *****
*****
ok: [192.168.56.106]

TASK [install samba package] *****
*****
changed: [192.168.56.106]

PLAY RECAP *****
*****
192.168.56.103      : ok=4   changed=0    unreachable=0    failed=0    skipped=2    rescue
d=0   ignored=0
192.168.56.104      : ok=4   changed=0    unreachable=0    failed=0    skipped=2    rescue
d=0   ignored=0
192.168.56.105      : ok=7   changed=1    unreachable=0    failed=0    skipped=3    rescue
d=0   ignored=0
192.168.56.106      : ok=6   changed=1    unreachable=0    failed=0    skipped=2    rescue
d=0   ignored=0

keliste@Kelliste-Workstation:~/CPE212_KELISTE$
```

Task 2:

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE
GNU nano 6.2 site.yml *
---
- hosts: all
  become: true
  pre_tasks:

    - name: install updates (CentOS)
      tags: always
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  tasks:

    - name: install apache and php for Ubuntu Servers
      tags: apache,apache2,ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      tags: apache,centos,httpd
```



```

GNU nano 6.2                                keliste@Keliste-Workstation: ~/CPE212_KELISTE
site.yml *
- php
  state: latest
  when: ansible_distribution == "CentOS"
- hosts: db_servers
  become: true
  tasks:
    - name: install mariadb package (CentOS)
      tags: centos, db, mariadb
      dnf:
        name: mariadb-server
        state: latest
        when: ansible_distribution == "CentOS"
    - name: "Mariadb- Restarting/Enabling"
      service:
        name: mariadb
        state: restarted
        enabled: true
    - name: install mariadb package (Ubuntu)
      tags: db, mariadb, ubuntu
      apt:
        name: mariadb-server
        state: latest
        when: ansible_distribution == "Ubuntu"
- hosts: file_servers
  become: true
  tasks:
    - name: install samba package
      tags: samba
      package:

```

[^]G Help [^]O Write Out [^]W Where Is [^]K Cut [^]T Execute [^]C
[^]X Exit [^]R Read File [^]\ Replace [^]U Paste [^]J Justify [^]/

```

keliste@Keliste-Workstation: ~/CPE212_KELISTE

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.105]

TASK [Mariadb- Restarting/Enabling] *****
changed: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
skipping: [192.168.56.105]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

TASK [install samba package] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.104      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.105      : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.106      : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$

```

Describe:

```

keliste@Keliste-Workstation:~/CPE212_KELISTE$ ansible-playbook -i hosts --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
    TASK TAGS: [always]

  play #2 (all): all    TAGS: []
    TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers    TAGS: []
    TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers TAGS: []
    TASK TAGS: [samba]

```

ansible-playbook --list-tags site.yml

Describe: This just lists every tag defined in the playbook without actually running any tasks

```
ok: [192.168.56.104]
ok: [192.168.56.103]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.104]
ok: [192.168.56.103]
ok: [192.168.56.106]
ok: [192.168.56.105]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.105]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103      : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.104      : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.105      : ok=6    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.106      : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

ansible-playbook --tags centos --ask-become-pass site.yml

Describe: This runs only the tasks tagged centos.

```
keliste@Keliste-Workstation: ~/CPE212_KELISTE

skipping: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.103]
ok: [192.168.56.104]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.104]
ok: [192.168.56.105]
ok: [192.168.56.106]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
skipping: [192.168.56.105]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103      : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.104      : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.105      : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.106      : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTES
```

ansible-playbook --tags db --ask-become-pass site.yml

Describe: This runs only the database-related tasks, skipping the web and file server tasks entirely.

```
TASK [Gathering Facts] *****
ok: [192.168.56.106]
ok: [192.168.56.104]
ok: [192.168.56.103]
ok: [192.168.56.105]

TASK [install apache and php for Ubuntu Servers] *****
skipping: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103 : ok=4 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
192.168.56.104 : ok=4 changed=1 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
192.168.56.105 : ok=5 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
192.168.56.106 : ok=5 changed=1 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$
```

ansible-playbook --tags apache --ask-become-pass site.yml

Describe: This runs only the apache/php installation tasks across web_servers, skipping the db and samba tasks.

```
ok: [192.168.56.104]
ok: [192.168.56.105]

TASK [install apache and php for Ubuntu Servers] *****
skipping: [192.168.56.105]
ok: [192.168.56.106]
ok: [192.168.56.104]
ok: [192.168.56.103]

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
skipping: [192.168.56.105]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103 : ok=4 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
192.168.56.104 : ok=4 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
192.168.56.105 : ok=6 changed=0 unreachable=0 failed=0 skipped=3 rescued=0 ignored=0
192.168.56.106 : ok=5 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
```

ansible-playbook --tags "apache,db" --ask-become-pass site.yml

Describe: This combines two tags in one run, so both the apache tasks and the db tasks execute together while file_servers tasks are skipped.

Task 3:

```
- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

```
100345 /usr/libexec/mariadb --basedir=/usr
[keliste-centos@KelisteMain CPE212_KELISTE]$ sudo systemctl stop httpd
[sudo] password for keliste-centos:
[keliste-centos@KelisteMain CPE212_KELISTE]$
```



Unable to connect

Firefox can't establish a connection to the server at 192.168.56.105.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to access the web.

[Try Again](#)

```

TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

TASK [start httpd (CentOS)] *****
skipping: [192.168.56.103]
skipping: [192.168.56.104]
skipping: [192.168.56.106]
ok: [192.168.56.105]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.105]

TASK [install mariadb package (CentOS)] *****
ok: [192.168.56.105]

TASK [Mariadb- Restarting/Enabling] *****
changed: [192.168.56.105]

TASK [install mariadb package (Ubuntu)] *****
skipping: [192.168.56.105]

PLAY [file_servers] *****

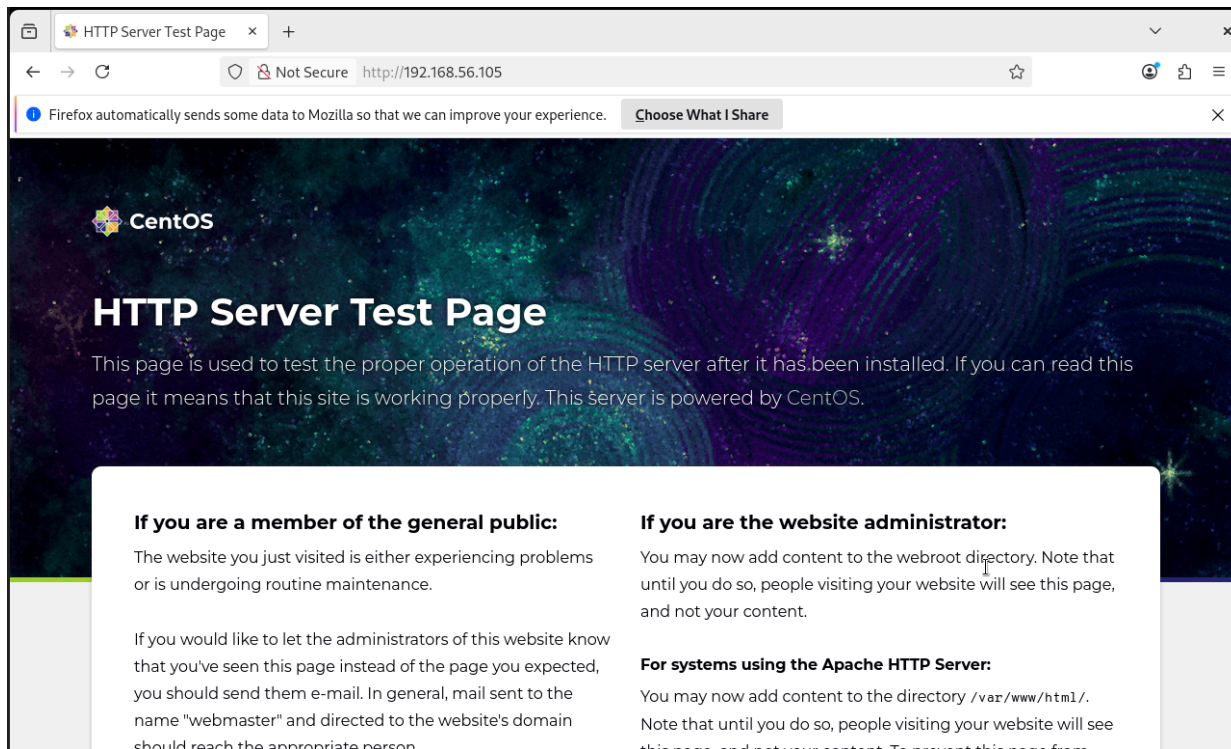
TASK [Gathering Facts] *****
ok: [192.168.56.106]

TASK [install samba package] *****
ok: [192.168.56.106]

PLAY RECAP *****
192.168.56.103      : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0
192.168.56.104      : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0
192.168.56.105      : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0
192.168.56.106      : ok=6    changed=0    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0

keliste@Keliste-Workstation:~/CPE212_KELISTE$

```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Creating a Role for Web Servers
2. Creating a Role for Web Servers
3. Creating a Role for Database Servers
4. Combining Multiple Roles in a Playbook

```

keliste@Keliste-Workstation:~/CPE212_KELISTE$ mkdir roles
keliste@Keliste-Workstation:~/CPE212_KELISTE$ /roles
bash: /roles: No such file or directory
keliste@Keliste-Workstation:~/CPE212_KELISTE$ cd /roles
bash: cd: /roles: No such file or directory
keliste@Keliste-Workstation:~/CPE212_KELISTE$ ls
hosts  install_apache.yaml  README.md  roles  site.yml
keliste@Keliste-Workstation:~/CPE212_KELISTE$ cd roles
keliste@Keliste-Workstation:~/CPE212_KELISTE/roles$ ansible-galaxy init roles/webrowsers
- Role roles/webrowsers was created successfully
keliste@Keliste-Workstation:~/CPE212_KELISTE/roles$ ansible-galaxy init roles/dbrowsers
- Role roles/dbrowsers was created successfully
keliste@Keliste-Workstation:~/CPE212_KELISTE/roles$
  
```

```

GNU nano 6.2                                roles.yml *
---
- hosts: web_servers
  become: true
  roles:
    - role: webrowsers
      tags: web

- hosts: db_servers
  become: true
  roles:
    - role: dbrowsers
      tags: db
  
```

Reflection:

1. What is the importance of putting our remote servers into groups?
 - Grouping remote servers matters because it lets me target exactly the machines a task applies to instead of running every play against every host.
2. What is the importance of tags in playbooks?
 - Tags let me run only the parts of a playbook I actually need at that moment, instead of re-executing the whole file every time.
3. Why do think some services need to be managed automatically in playbooks?
 - Some services don't start automatically after installation because the package manager only installs the software itself and it doesn't assume I want it running immediately.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.